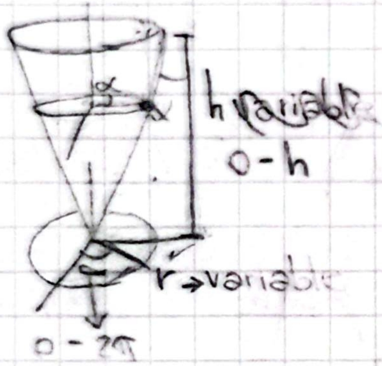
Problemas viernes 21 de Agosto:

1. Calcule la trayectoria de la distancia mínima entre 2 puntos sobre la superficie de un cono invertido con ángulo en el vértice α.

Recordemos cilíndricas:



$$dR = \langle x, y, z \rangle$$

$$dR = \langle r\cos\theta, r\,\text{sen}\,\theta, z \rangle$$

Para un cono: $z = kr$

$$\rightarrow dz = k\,dr \Rightarrow dz^2 = k^2 dr^2$$

$$\Rightarrow ds^2 = dr^2 + r^2 d\phi^2 + dz^2$$

$$\wedge\ ds^2 = dr^2 + r^2 d\phi^2 + k^2 dr^2 = (1+k^2)dr^2 + r^2 d\phi^2$$

Para hallar la mínima distancia:

$$I = F(y(x)) \rightarrow y(x) = r(\phi) = r \rightarrow dr = \frac{dr}{d\phi}\,d\phi \rightarrow dr = r'\,d\phi$$

$$\rightarrow I = \int_{\phi_1}^{\phi_2} r(\phi)\,d\phi = \int_{\phi_1}^{\phi_2} \sqrt{(1+k^2)r'^2 + r^2}\,d\phi$$

2. Calcule el valor mínimo de

$$I = \int_0^1 \left[(y')^2 + 12xy\right] dx \quad ; \quad y(0)=0 \;\;\text{y}\;\; y(1)=1$$

$$\rightarrow I = \int_0^1 \left(\frac{dy}{dx}\right)^2 dx + \int_0^1 12xy\, dx$$

Problemas 28 Agosto

Goldstein

① $\mathcal{L} = \frac{1}{2}a(\dot{x}^2 \operatorname{sen}^2 y + \dot{y}^2) + \frac{1}{2}b(\dot{x}\cos y + \dot{z})^2$

a) $\frac{d}{dt}\left[\frac{\partial}{\partial \dot{x}} \frac{1}{2}a\left(\dot{x}^2\operatorname{sen}^2 y + \dot{y}^2\right) + \frac{1}{2}b\left(\dot{x}\cos y + \dot{z}\right)^2\right] - \cancel{\frac{\partial}{\partial x}\left(\mathcal{L}\right)}^{0} = 0$

$\rightarrow \frac{d}{dt}\left[\frac{1}{2}a\left(2\dot{x}\operatorname{sen}^2 y\right) + \frac{1}{\cancel{2}}b\,\cancel{2}\left(\dot{x}\cos y + \dot{z}\right)\cos y\right]$

Para x

$\rightarrow \frac{1}{2}a\left[2\ddot{x}\operatorname{sen}^2 y + 4\dot{x}\cos y\right] + b\left[\left[\ddot{x}\cos y - \dot{x}\operatorname{sen} y + \ddot{z}\right]\cos y - (\dot{x}\cos y + \dot{z})\operatorname{sen} y\right] = 0$

Para y:

$\rightarrow \frac{d}{dt}\left(\frac{\partial}{\partial \dot{y}}\left[\mathcal{L}\right]\right) - \frac{\partial}{\partial y}\left[\mathcal{L}\right] = \frac{d}{dt}\left[\frac{1}{\cancel{2}}a\cdot\cancel{2}\dot{y}\right] - \frac{1}{\cancel{2}}a\left(\cancel{2}\dot{x}^2\cos y \operatorname{sen} y\right) - b\left(\dot{x}\cos y + \dot{z}\right)\left(-\dot{x}\operatorname{sen} y\right)$

$a\ddot{y} - a\left(2\dot{x}\operatorname{sen} y + \dot{x}^2\cos y\right) - b\left[\left(\ddot{x}\cos y - \dot{x}\operatorname{sen} y + \ddot{z}\right)\left(-\dot{x}\operatorname{sen} y\right) + \left(\dot{x}\cos y + \dot{z}\right)\left(-\ddot{x}\operatorname{sen} y + \dot{x}\cos y\right)\right]$